

GRADUATION PROJECT

(END-TO-END DATA ANALYSIS USING SQL
SERVER & MICROSOFT POWER BI)

ROUTE

01 | DATA SOURCE

[Datasets Link \(Click Here\)](#)

02 | CONTEXT

You are working as a Data Analyst for a Retail Company operating multiple stores across regions.

The company collects data from different systems and years, but decision makers lack a unified analytical view.

Management wants:

- One trusted data model
- Clear KPIs
- Interactive Power BI dashboards
- Actionable business insights

Your task is to design, analyze, and visualize this data using SQL Server and Power BI, applying everything you learned in the diploma.

REQUIRED SUMMSION FILES

- Presentation File shows all analysis stages
- Interactive Power BI report
- Sql server backup
- A documented SQL file containing all queries

REQUIRED TASKS

Graduation Project Roadmap

Phase 1: Data Understanding & Plan

Review all CSV files to:

- Understand the business problem and put business questions
- Identify facts and dimensions
- Plan the data model

Phase 2: SQL Server – Database Design & Loading

- **Create a new database in SQL Server**
- Create tables with proper:
 - Data types
 - Primary keys
 - Foreign keys
- Load data from CSV files into SQL Server

Phase 3: SQL Analysis & Business Queries

- Answer business questions you put by using SQL(joins, aggregations, and filters)

Phase 4: SQL Views & Data Preparation

- Prepare analysis-ready data for Power BI (SQL Views)

Phase 5: Power BI – Data Modeling

- Connect Power BI to SQL Server
- Load fact and dimension tables and manage your Relationships
- Build a clean semantic model

Phase 6: DAX Measures & KPIs

- Apply DAX for business metrics like:
 1. Total Sales
 2. Total Orders
 3. Average Order Value
 4. Total Returns
 5. Return Rate %
 6. Year-over-Year Growth
 7. Month-over-Month Growth

REQUIRED TASKS

Phase 7: Power BI Dashboard Design

- Use Figma to desing dashboard professional frame
- Communicate insights visually
- Apply data storytelling principles (at least 3 report pages in addition to home Page) ex) Executive Overview, Sales & Product Analysis, and Customer & Regional Insights.

Phase 8: Business Insights & Recommendations

Prepare a final Insights Summary answering:

1. What drives sales growth?
 2. Which products or regions underperform?
 3. What is the impact of returns?
 4. Where should management focus next quarter?
- **Each insight backed by data from the dashboard**

Move from numbers to decisions

Project Evaluation

